

John Gargalionis

johngargalionis@gmail.com • www.johngargalionis.com • +61-418-637-727 • Citizenship: Australian • LinkedIn: [john-gargalionis](#)

Sections below prefixed 'Selected' have been abridged. Please see [my complete academic CV](#) for the full lists of 7 publications, 13 talks, 7 awards, 10 teaching positions and a summary of my scientific outreach. My publications and associated bibliometrics can also be accessed at my [Inspire HEP](#) and [Google Scholar](#) profiles. Professional references can be provided upon request.

Education

- 2016 – 2020 **The University of Melbourne**
PhD in Theoretical Particle Physics
Advisor: Prof. Raymond Volkas
Thesis: [Models of radiative neutrino mass and lepton flavour non-universality](#)
- Doctoral work awarded [Chancellor's Prize for Excellence in the PhD](#)
 - Lead author on 5 scientific articles with a total of 221 citations as of Sep 2022
 - Presented work at 12 talks, including 2 invited lectures at [CERN](#) and [KEK](#)
- 2014 – 2016 **The University of Melbourne**
MSc in Theoretical Particle Physics (First Class Honour)
Advisors: Prof. Raymond Volkas & Prof. Elisabetta Barberio
Thesis: Neutrino mass through leptiquarks: a new radiative model and its experimental prospects
- Coursework included studies of [non-equilibrium statistical mechanics](#)
 - Received the Prof. Kernot Scholarship in Physics (highest graduating mark)
 - Received the N. D. Goldsworthy Scholarship for excellence in graduate coursework
- 2010 – 2014 **The University of Melbourne**
BSc with a Physics major (First Class Honour)
- Additional specialisations in Classics (Ancient Greek and Latin) and Neuroscience

Experience

- Jan 2022 – Present **Investigador Doctor Senior (IFIC and the University of Valencia, Spain)**
Leading a research programme in automated model building and phenomenology
- Successfully graduated an MSc student under my supervision
 - Awarded the [Juan de la Cierva Fellowship](#) (two years' funding for my own personal research programme)
- May 2021 – Oct 2021 **Research Associate (The University of Melbourne)**
Worked on novel particle-identification techniques using unsupervised machine learning and data from the Large Hadron Collider

Selected teaching experience

- Oct 2022 **Invited discussion leader:** [Asia-Europe-Pacific School of High Energy Physics \(South Korea\)](#)
Relevant topics include lectures on statistics
- Jan 2018 **Tutor:** [Advanced Scientific Programming in Python Asia-Pacific School \(Melbourne\)](#)
Included topics: version control using Git; Python parallelism; context managers and generators; data cleaning and visualisation; Cython, numba and code profiling
- 2014 – 2018 **Tutor, [PHYC20013](#) and [PHYC30021](#):** **Computational Physics (The University of Melbourne)**
Taught high-performance scientific computing, numerical methods and data analysis in C and C++

Technical skills

- Programming languages** Proficient in:
- Python (featured project: [neutrinomass](#))
 - Mathematica (featured project: [PythonForm](#))
- Familiar with:
- Haskell (featured project: [MuonAnomalyModels](#))
 - C++ (featured project: [CSVProducer](#))
 - Clojure and other Lisps (featured project: [completions](#))
- Unlinked projects available upon request*
- Libraries and frameworks**
 - Scientific Python: NumPy, SciPy, SymPy, Pandas, Jupyter, *etc.*
 - High-quality data visualisation with Matplotlib, Seaborn and Plotly
 - Machine learning with TensorFlow and Keras
 - $\text{\LaTeX}$
 - Git
 - At home at the Unix command line
 - Experienced in cluster computing and job submission with PBS and slurm
- Scientific analysis**
 - Expert in building mathematical models to explain data and make predictions
 - Proficient in Monte-Carlo simulations and statistical analyses
- Communication**
 - Public speaking (see [invited talk at CERN](#) from 2017, or [recent seminar](#) in Korea)
 - Scientific writing (see [PhD thesis](#); preface is low on technical jargon)
- Languages** English (native), Modern Greek (fluent), Spanish (intermediate)

Other interests

I enjoy chess, languages and pizza.